

RISTEK Hackathon 2026 - Preliminary Proposal



WaveGuard

Extending Home Security Powered by Wi-Fi Sensing and AI.
No Blind Spot, Privacy Protected, This is WaveGuard.

by 

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1. Executive Summary

For dual-income households whose homes sit unattended for hours each day, WaveGuard is a WiFi sensing system that extends home security into the rooms a camera will never enter, unlike CCTV, which protects only the spaces a family is willing to expose to a lens.

The system turns a household's existing WiFi signal into a motion sensor. Four ESP32-S3 nodes mounted on interior walls read how a moving body perturbs that signal, while a fifth node fuses their readings into a single floor-plan view of activity across the home. WaveGuard reports presence at the zone level regardless of how many people are inside, and resolves approximate position for up to three simultaneous movers by comparing signal strength across the node mesh. Only a compact feature summary leaves each node rather than the raw signal itself, so the privacy claim comes from what the hardware is able to transmit, not from a policy a company could later change.

This gap matters because Indonesia's urbanization is pushing more households into the exact pattern of daytime absence that property crime data flags as a burglary marker, and the CCTV these households already rely on carries a documented failure rate above fifty percent against local theft, with a hard limit at the private rooms no camera will ever cover. WaveGuard does not replace that camera. It is the layer that watches where the camera cannot, giving a family checking their phone from work an honest answer about the house they left behind.

2. Problem Statement

2.1 Background & Context

Indonesia is undergoing one of the fastest urbanization trajectories in Southeast Asia: Badan Pusat Statistik data shows 56.7 percent of the population was urban by 2020, projected to reach 66.6 percent by 2035 as economic activity concentrates in hubs like Jakarta. This concentration carries a documented social cost, with rising criminality disproportionately exposing the commercial and residential assets clustered in these

urban centers.

As cities absorb this influx, CCTV has become the default security response, yet this reliance has gone largely unexamined. Fixed optical hardware leaves architectural blind spots installers cannot always predict, and stops at a social boundary: private rooms such as bedrooms will never be fitted with a camera no matter how advanced the hardware becomes. The following section quantifies this protection gap using criminal-justice data and hardware-limitation evidence.

2.2 Data & Evidence

This section validates the security gap in traditional CCTV using national crime data and physical device limitations.

A. Sectoral Property Crime in Indonesia

Police data from the National Criminal Information Center ([\(Pusiknas\) Bareskrim Polri](#)) shows that Theft with Infiltration, break-ins targeting commercial offices, empty warehouses, apartments, and private facilities, comprises 11.7% of all recorded criminal indictments in Indonesia. Tactical forensic assessments from the same source show tens of thousands of these crimes deliberately target facilities protected only by passive optical infrastructure with operational blind spots, primarily unoccupied warehouses, offices, and rukos outside working hours. [Kriminalitas properti](#)

B. Technical Limitation of the CCTV

Despite the dense implementation of optical surveillance networks, empirical benchmarks reveal these hardware still have some fatal vulnerabilities:

- 1. Crime Prevention Failure Rate:** According to specialized security systems evaluations published in the [Jurnal JIFORTY Universitas Bhayangkara Jakarta Raya](#), standard commercial CCTV networks only achieve a maximum **47.4% reduction in local property theft**. Concurrently, this leaves a severe **52.6% protection gap** where intrusions go undetected or unprevented.
- 2. Technical Limitations:** Traditional CCTVs carry three structural weaknesses: static lenses restrict the field of view to 90°–110° and leave a 15°–25° blind spot

beneath the camera, exposed lenses are easily blocked or vandalized, and object-classification accuracy drops 40%–60% under dim-light conditions.

3. Residential Vulnerability: [Pusiknas Bareskrim Polri](#) further reports that empty house burglary is not confined to luxury properties: perpetrators watch for signs like an absent vehicle before striking, a pattern that peaks when every adult in a household holds a daytime job.

2.3 User / Persona

WaveGuard's primary persona is Rani, a 34-year-old marketing manager in South Tangerang whose household matches the exact daytime-absence signal Pusiknas flags as a burglary marker: both parents commute full-time, leaving the house empty each weekday.

Rani installed CCTV at the gate and living room after a neighbor's break-in, but coverage stops at her children's bedroom door, a boundary she sets herself rather than a technical limit. The app can only confirm what already happened in the rooms it watches; it offers nothing about the hours the house sits unattended.

That risk carries a real price: one reported empty-house burglary in 2026 left the homeowner facing losses of thirty million rupiah, discovered only after a short errand. Rani has no way of knowing whether her own absence carries the same cost until it already has.

3. Proposed Solution: WaveGuard

3.1 Solution Overview

WaveGuard is the missing layer for a household like Rani's, closing the exact gap her CCTV cannot reach without asking her to give up the boundary she already set for her own family. Four ESP32-S3 nodes mounted on interior walls turn the home's existing WiFi signal into a motion sensor, with no camera added anywhere in the house. This is not experimental guesswork: WiFi-based device-free sensing has been an active academic research field for over a decade, with intrusion detection and smart home monitoring among its established applications. WaveGuard applies that research to a

mesh of four wall-mounted nodes reporting to one hub, reporting zone-level presence and approximate position (detailed in Section 4) while deliberately declining to count occupants or identify who moved, since a product that claimed more precision than a WiFi reflection can support would be lying to Rani exactly when she needs the truth most.

3.2 System Workflow

The human body is largely water, which reflects 2.4 GHz radio waves, so a moving person perturbs the signal at a receiver. Four ESP32-S3 nodes on the walls measure that perturbation and a fifth fuses their readings, so the system runs on five microcontrollers with no computer and no camera. Figure A1 (Appendix A) shows the five stages end to end.

CSI exists only when a packet arrives, and at night nobody uses the Wi-Fi, so each node solicits the router one hundred times per second and measures CSI on the reply. This creates a binding prerequisite: every node must measure frames from the same router. After on-node processing only a 120-byte feature vector leaves the device at 20 Hz over ESP-NOW, so raw CSI never crosses that boundary and the privacy claim is architectural rather than a policy statement. Channel and BSSID stay locked, since a change invalidates calibration and breaks ESP-NOW, and clock skew stays below 20 ms.

The master fuses the readings into positions and matches them against floor-plan zones. The trigger policy never counts people: it compares activity against the occupancy the user declares when arming Alert Mode, so if the home is declared empty and activity appears anywhere, an alert is raised. This turns an unsolvable problem into a comparison against a known number, and motion to decision takes about 0.1 s.

4. Technical Architecture & Tech Stack

4.1 System Architecture

Sensing (L1) is the existing router plus four wall-mounted ESP32-S3 nodes emitting raw CSI at 100 Hz each. Edge processing (L2) runs inside every node, reducing that to a

120-byte feature vector at 20 Hz. Transport (L3) is ESP-NOW between chips on the locked channel, with no broker. The master (L4), a fifth ESP32-S3, performs multilateration, zone mapping, classification, calibration, and policy within 3 percent of its compute budget. The application (L5) presents the result. Figure A2 (Appendix A) lays this out. The boundary between L2 and L3 is the load-bearing decision: because each node emits only a nine-number summary, raw channel data has no path off the device at all. Everything through L4 runs inside the home and keeps working without internet; only sign-in and remote push need an outbound connection.

4.1.1 Ranging and multilateration

Position is estimated by comparing signal-strength ratios across pairs of nodes rather than reading any single node in isolation, since this ratio cancels out the unknown factors of body size and router distance (full derivation in Appendix B). Four nodes are the minimum needed to resolve a single position without ambiguity. Occupants moving at different radial velocities are separated using Doppler bins, resolving up to three simultaneous targets before the estimate falls back to zone-level presence. Simulation on a 20 cm grid gives 0.8 to 1.5 m RMSE within a room and 2 to 3 m through walls, with zone correctness above 90 percent.

4.1.2 Human versus disturbance classification

Wi-Fi alone cannot tell a person from a fan, a curtain, or a pet, so this is where machine learning earns its place. A compact classifier runs on the master over three-second windows, reading distinct signatures: a fan is periodic and narrowband and visible to one node only, a curtain is slow and stationary, a pet reflects less power and moves erratically, and a human is aperiodic, broadband, and migrates between nodes. Training data is recorded and labelled by the team, with users later flagging false alarms in the app to refine it. Positioning itself is deliberately not learned, since a model trained in one home fails in the next while the physics does not.

4.2 Signal Processing Approach

Every WaveGuard sensor reads WiFi signal data the router already sends out, using a well-established technique from prior research [3], [7], [8]. It keeps only how strong the signal is, not its exact timing, because on this hardware the timing data gets scrambled in a way that can't be fixed without an extra piece of equipment none of the sensors have. That rules out fancier tracking methods used in some research systems [4], so WaveGuard instead relies on comparing signal strength across sensors, the math for which is laid out in Appendix B.

Each sensor cleans up its reading in four steps: correcting for an automatic volume-control effect in the chip, filtering out noise spikes, discarding unreliable frequency channels, and boiling everything down to a single number that rises and falls as something moves nearby. That number gets sliced into one-second chunks to detect motion, shown step by step in Figure A3a. Only this compressed motion summary ever leaves the sensor, not the detailed radio data, so the finer details a system would need to do things like watch specific gestures are gone before they ever leave the device. That's the actual privacy guarantee, not a rule someone could change later.

Machine learning is used in exactly one place: telling a person apart from a fan, curtain, or pet, done by the classifier described in Section 4.1.2. Figuring out where someone is standing doesn't use machine learning at all; it's solved with straightforward geometry. That's deliberate. The geometry works the same in any house, so it only needs a brief one-time setup per home. A learned model wouldn't transfer as cleanly, which is a known problem in similar research [1], [5].

4.3 Technology Stack

Five ESP32-S3-WROOM-1 units, four nodes plus one master, work alongside the existing router and the user's phone. Firmware is ESP-IDF v5.2 LTS, bundling FreeRTOS, ESP-NOW, NVS, and mbedTLS, with espressif/esp-csi for acquisition and ESP-DSP for FFT, all permissively licensed; the app is Flutter for both platforms.

ESpectre is deliberately avoided, since its GPLv3 licence would force disclosure of the entire firmware source.

4.4 Feasibility & Validation Plan

The accuracy figures above come from modelling and literature, not our own measurements. Two results are verified numerically: the additivity of power ratios, why four nodes are required, and a grid-likelihood simulation that resolved two simultaneous targets within one 20 cm cell; three-target separation remains to be measured. Validation proceeds in four steps: AGC compensation (R^2 above 0.9, binding since a wrong compensation invalidates all ranging), single-target multilateration below 1.5 m RMSE over nine known points, two-target separation within 2 m, and end-to-end latency under 0.2 s.

Five weaknesses are acknowledged: targets sharing a radial velocity or exceeding three simultaneous movers fall back to per-room presence; occupants are never counted, only reported as room activity; a still target shows its last known zone; false alarms are held down only by Alert Mode's declared-occupancy check until the classifier is trained; and no cross-home field trial has been run, our largest risk. Pose reconstruction, individual identification, and heartbeat detection are not claimed.

5. Core Features & User Journey

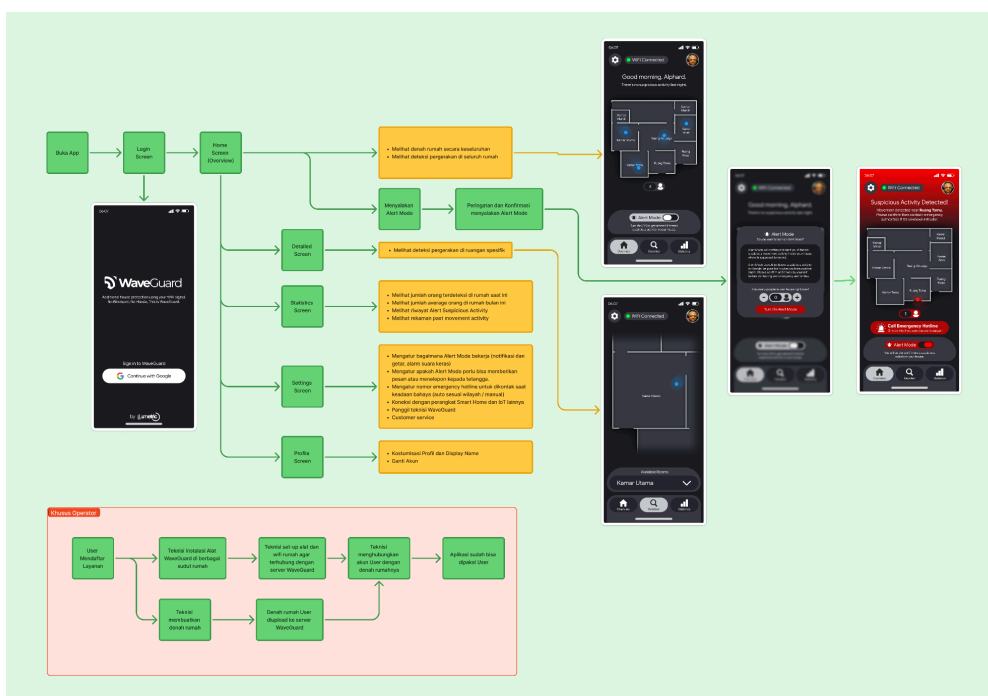
5.1 Core Features (MVP Scope)

The MVP covers four core capabilities, each scoped to what four wall-mounted nodes and one master can support without a camera, a computer, or a cloud dependency: zone-level presence (Section 3.1) with no cap on occupant count; multilateration (Sections 3.1, 4.1.1) tracking up to three simultaneous movers; Alert Mode (Section 3.2) comparing activity against declared occupancy; and an on-device human-versus-disturbance classifier (Section 4.1.2), the single MVP feature still in training. All sensing, fusion, and alert decisions run inside the home, with raw CSI never leaving a node. Occupant counting, identity recognition, pose reconstruction,

vital-sign sensing, camera integration, multi-floor coverage, and automatic node self-survey all sit outside MVP scope.

5.2 User Journey

WaveGuard's user journey begins at sign-in, where the resident authenticates via Google and lands on the Overview screen: a floor-plan view showing zone-level presence and the current occupant count, with no per-user identity attached to any dot on the map. From there the resident chooses whether to arm Alert Mode. Arming asks a single question, how many people remain in the house, and that declared number becomes the baseline every subsequent alert is checked against (Section 3.2). With Alert Mode on, the app returns to a quiet Overview state until the master node detects activity outside the declared occupancy. When that happens, the resident receives a push notification naming the zone where movement was detected and is prompted to confirm the activity or, if it can't be explained, contact the emergency hotline. Every alert can be flagged as a false alarm directly in the app, closing the loop between what a resident experiences and what the human-versus-disturbance classifier in Section 4.1.2 later learns to filter.



6. Innovation & Uniqueness

6.1 Comparison to Existing Solutions

Category	Key Limitation	Multi-Room Coverage
CCTV	Line-of-sight dependent, stops at private-room boundary, vulnerable to physical tampering	Independent cameras, no shared spatial map
PIR Motion Alarm	False triggers from heat sources (windows, kitchens), narrow sensing cone	Multiple independent units, no shared floor plan
Radar/mm Wave	No heat dependency, but still single-node	Several independent sensors, no fusion
WaveGuard	Penetrates walls, four-node mesh fused into one hub	One hub fuses four nodes into a single floor-plan view

None of the three existing categories fuse multiple sensing points into a shared spatial output; that gap is the space WaveGuard is built to close.

6.2 Differentiation

WaveGuard's differentiation is structural, not incremental: single-node categories in the table above cannot fuse a shared spatial view because ratioing position requires a second node to ratio against (Section 4.1.1). The four-node mesh is what turns isolated motion triggers into one floor-plan read, the gap none of the three existing categories close.

7. Business Model & Sustainability

7.1 Business Model Canvas Summary

[Link BMC](#)

7.2 Target Market & Customer Segment

- **Primary segment:** dual-income residential households in urban and high-density areas, matching the daytime-absence pattern identified in Section 2 as a burglary marker.

- **Secondary segment:** privacy- and security-conscious individuals with a smart-home mindset, unwilling to install cameras in private rooms.

7.3 Revenue Stream

WaveGuard's revenue model combines primary and secondary income. Primary income comes from a one-time hardware bundle, four pre-configured ESP-32 nodes plus a free one-month software trial, followed by a monthly software subscription after the trial ends. Secondary income comes from spare part sales: individual pre-programmed ESP-32 replacement units for long-term hardware uptime

7.4 Go-to-Market Strategy

WaveGuard's go-to-market runs in three phases:

- Phase 1: Validation & Grassroots (Months 1-6) : validate performance against the metrics in Section 8.2, build early awareness through social media and Indonesian tech reviewers, and pursue an ISP as a strategic partner
- Phase 2: Commercialization (Months 7-12) : promote through residential forums with a privacy-first message, offering a complimentary one-month trial with every initial bundle, focusing commercialization in jabodetabek.
- Phase 3: Mass Scaling and Expansion (Year 2+) :Expanding sales nationwide, partner with urban property groups to embed WaveGuard into default building blueprints, extending the zone-level scope defined in Section 9.1.

8. Expected Impact

8.1 Social / Industry Impact

WaveGuard's social impact follows directly from the persona in Section 2.3: as urbanization pushes more households into daytime absence, four wall-mounted nodes close CCTV's blind spot in private rooms without adding a lens, and because raw CSI never leaves a node, that privacy guarantee is architectural rather than a policy a company could change later. Its industry impact lies in the path from hackathon prototype to distributable product: built on a permissively licensed firmware stack, WaveGuard can pursue regulatory certification and integration with ISP and real-estate

infrastructure, giving Indonesian households device-free sensing without new cabling or dedicated hardware

8.2 Success Metrics

The successful development and implementation of this ESP32-based Wi-Fi Sensing technology is not measured partially, but evaluated through an integrated framework known as the Model-Product-Business Metric Chain.

A. Model Metrics

- Zone accuracy (Target: $\geq 90\%$): the share of events where the reported room matches ground truth, measured over nine marked positions.
- Position RMSE (Target: ≤ 1.5 m in-room): median error for a single mover, degrading to 2 to 3 m through walls.
- Human versus disturbance F1 (Target: ≥ 0.90 on a held-out set): measured against a labelled recording of walking, fans, curtains, and pets. This model is not yet trained; the figure is a target, not a result.
- False alarms per armed home per week (Target: ≤ 1): a count rather than a ratio, because a single nightly false alarm is enough for a user to disable the system regardless of percentage.

B. Product Metrics

- Motion to decision (Target: ≤ 0.2 s): measured on the master node, independent of any network.
- Decision to phone notification (Target: ≤ 5 s): dominated by the push relay and the user's mobile network, not by our processing.
- Installation completion (Target: $\geq 90\%$): the share of users who finish node placement and calibration without support contact.
- False-alarm flag rate: the share of alerts users mark as false in the app, which both measures quality and supplies training labels for the classifier.

C. Business Metrics

- Homes deployed (Target: 5,000 in year one): counted as completed installations, concentrated in high-incidence areas of Jabodetabek.
- Churn (Target: $\leq 2\%$ per month): roughly 22% annually, appropriate for a hardware-anchored product where the device stays in the home.

9. Roadmap & Future Development

9.1 Long-Term Vision

WaveGuard's long-term vision is to become the default privacy-preserving sensing layer for Indonesian homes and small businesses, scaling the MVP's honest zone-level foundation rather than replacing it. Near-term extensions are multi-floor coverage, coordinating more than one master node, and automatic node self-survey, replacing the manual calibration walk in Section 4.1.1. Further out, the same shape-reading classifier can be trained to recognize other motion patterns worth flagging, most notably a fall, without adding capabilities the architecture was deliberately built to avoid, such as identifying who is in the room or reading a vital sign. On the business side, the ISP partnership from Section 7.4 extends into full software bundling alongside the real-estate integration in Phase 3, so device-free sensing becomes something a household inherits rather than seeks out on its own.

References

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Appendix

Appendix A. End-to-End System Workflow

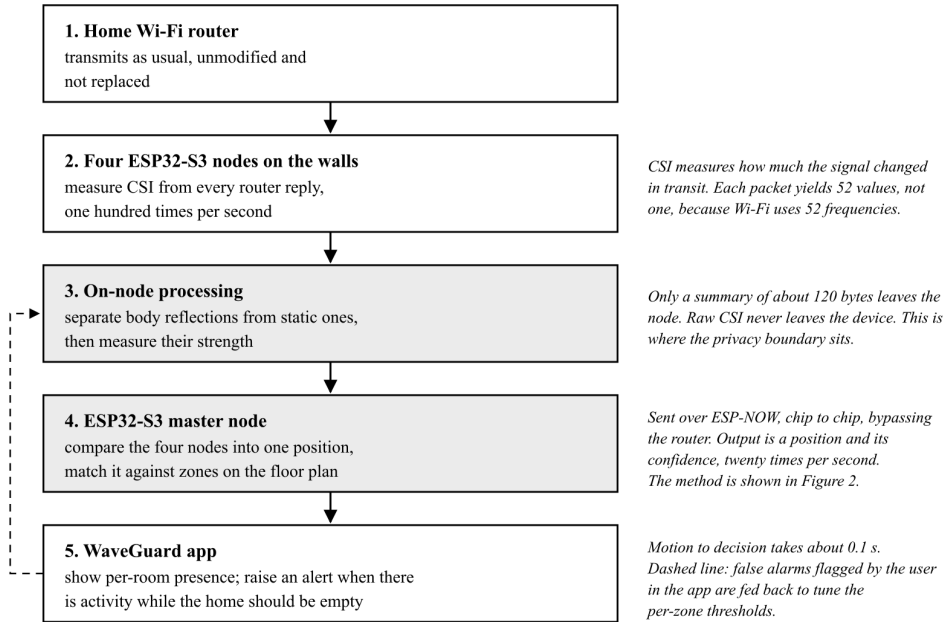


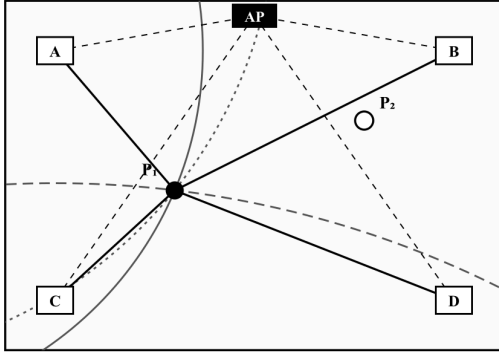
Figure A1. System workflow and division of responsibility. Steps 1 to 3 run inside each sensing node, step 4 on the master node, and step 5 in the mobile application. The privacy boundary lies between steps 3 and 4: only a 120-byte summary crosses it, never raw CSI.

Layer	Contents	Output and rate
L1 Sensing	existing home router, unmodified, plus four wall-mounted nodes	raw CSI, 100 Hz per node
L2 Edge, in node	AGC compensation, static and dynamic split, range estimation	120-byte vector, 20 Hz
Trust boundary: raw CSI never crosses this line, only a nine-number summary		
L3 Transport	ESP-NOW, chip to chip on the locked channel, no broker	feature stream, 20 Hz
L4 Master node	multilateration, zone mapping, classifier, calibration, policy	position, under 3 % of CPU
L5 Application	presence map on the floor plan, Alert Mode, notifications	user alerts, per event

Figure A2. Layer decomposition. Each layer is testable and replaceable on its own, and the trust boundary between L2 and L3 is enforced by what the firmware transmits, not by policy.

Appendix B. Ranging Geometry and Solver Output

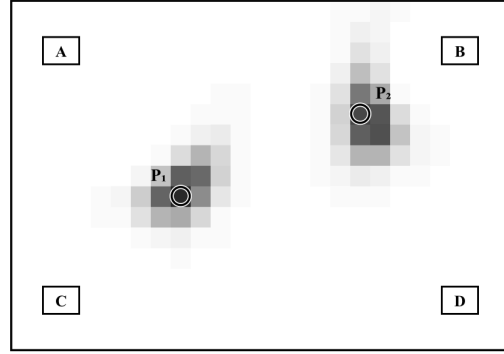
(a) Geometric constraints: Apollonius circles



Same scene as (b). Only the constraints of P_1 are drawn, and they meet at one point.

Both panels show the same $5.0 \text{ m} \times 3.5 \text{ m}$ floor plan at the same instant, with two people present; height is not estimated. Because P_1 and P_2 move at different radial velocities they fall into separate Doppler bins, and within a bin only one of them contributes, so the ratio identity is applied per bin exactly as it is for a single person. Panel (a) draws the constraints of P_1 alone, purely to stay legible; P_2 is solved the same way from its own bin. Panel (b) is the combined output, and both peaks land within one cell, 6 cm, of truth.

(b) Solver output: 2-D likelihood map



Two targets resolved, one per Doppler bin. Darker means a higher likelihood.

Figure B1. Apollonius constraints and the 2-D likelihood map. Both panels show the same instant with two people present. Panel (a) draws the constraints of P_1 alone to stay legible; P_2 is solved identically from its own Doppler bin. Panel (b) is the combined solver output.

The signal travels router to body to node, with the body as a scatterer. The bistatic radar equation gives the power at node i , where c_i is that node's link constant, σ the target radar cross-section, d_t the router-to-target distance, and d_i the target-to-node distance:

$$P_i = C_i \cdot \sigma / (d_t^2 \cdot d_i^2)$$

Both σ , which depends on body size and orientation, and d_t are unknown and will never be known, so a strong reflection may mean a small person nearby or a large person far away. Inter-link ratios cancel both. Since all nodes measure the same router, d_t is identical at any instant and σ is likewise shared, so both vanish when two links are divided:

$$\gamma_{ij} = d_i / d_j = \sqrt{((C_i / C_j) \cdot (P_j / P_i))}$$

Only the ratio c_i/c_j then needs calibrating, obtained by having the user move in place at a few marked points; absolute scale becomes unobservable, but position does not. Points whose distance ratio to two fixed nodes is constant form an Apollonius circle, $|x - s_i| = \gamma_{ij} |x - s_j|$. Four nodes are the minimum, not three: because $\ln(d_i/d_j)$ is additive, n nodes

give only n minus 1 independent constraints, so three nodes leave two circles meeting at two points, and simulation places that ghost 2.19 m from truth, inside the room. Figure B1a (Appendix B) draws the construction.

Occupants moving at different radial velocities fall into different Doppler bins, and within a bin one target dominates, so the ratio identity applies per bin unchanged. The solver searches jointly over position and velocity at 0.06 m/s resolution, and peaks in that space are targets. Up to three simultaneous targets are resolved this way, as Figure B1b demonstrates for two; two people moving at the same radial velocity collapse into a single bin, and beyond three the joint search can no longer separate them cleanly. In both cases the system falls back to per-room presence, which carries no target limit. Scoring a 20 cm grid gives 0.8 to 1.5 m RMSE within a room and 2 to 3 m through walls, with zone correctness above 90 percent.

Appendix C. UI Mockup

